

Vaishvi Patel

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EDUCATION

Master of Computer Science, North Carolina State University Aug 2025 - May 2027
Relevant Coursework: Algorithms, Software Engineering, Data Analysis, AI of Things, Computer Networks (**Graduate Teaching Assistant**), Neural Network, Deep Learning, Parallel Systems, ML with graphs, Gen AI GPA: 3.9/4.0
Bachelor of Engineering, Computer Science, Gujarat Technological University Aug 2021 - Jun 2025
New L.J. Institute of Engineering and Technology CGPA: 9.8/10.0

SKILLS

Languages & Web	Python, Java, C, JavaScript, SQL, HTML, CSS, React, Node.js
Backend & Cloud	Flask, FastAPI, REST API, AWS (S3, EC2, Rekognition), Docker, Git, CI/CD
Data	Kafka, Airflow, dbt, DuckDB, Parquet, Streamlit, PostgreSQL, MongoDB, Pandas, NumPy
AI/ML	PyTorch, TensorFlow, Keras, Scikit-learn, NLP, BERT, VADER, TextBlob
GenAI & LLMs	LLM Serving, KV-Cache Optimization, RAG, Prompt Engineering, LLM APIs

EXPERIENCE

Graduate Research Assistant | [Robotics Lab](#) | LLM, KV-Cache, ML, Data Analysis May 2026 - Aug 2026
NC State University - Dept. of Computer Science, advised by Prof. Khaled Harfoush *Raleigh, NC*

- Characterized LLM workload signatures (output-length distributions, prefix-reuse patterns) across 5,000+ requests from LongBench and HumanEval traces, finding 40% of prompts share reusable prefixes; results informed data-driven scheduling policies.
- Built an ML-based scheduling policy for KV-cache memory management in a full-stack LLM serving system, improving P95 latency by 18% over a FCFS baseline on multi-tenant workloads.

Software Engineer Intern | AWS Rekognition, Computer Vision, Data Pipelines Jan 2025 - Apr 2025
Cignex Technologies *Ahmedabad, India*

- Delivered a sketch-to-photo recognition system on AWS Rekognition with 85% accuracy across 500 samples.
- Enhanced model precision by 15% via histogram equalization and optimized data pipelines to cut latency by 20% across 1,000 image pairs.

Artificial Intelligence Intern | IBM Watson, Python, Flask, NLP Jun 2024 - Jul 2024
IBM SkillsBuild Program with CSRBOX *Ahmedabad, India*

- Built an AI movie recommendation app using IBM Watson, Python, and Flask achieving 88% user satisfaction.
- Refined NLP-based relevance by 30% and reduced response time by 25% (4s → 3s).

PROJECTS

Real-Time Data Engineering Pipeline | Python, Kafka, Airflow, dbt, DuckDB, Parquet, Streamlit, Docker

- Built end-to-end data pipeline (Hacker News API → Kafka → Python → Parquet medallion lakehouse → dbt + DuckDB → Streamlit) orchestrated by Airflow with 15-min dbt triggers and data quality gates; chose DuckDB over a hosted warehouse for local-first analytics at this scale, avoiding infrastructure overhead.
- Deployed ensemble NLP sentiment analysis (VADER + TextBlob) scoring 500+ posts/hour; containerized with Docker Compose and CI/CD via GitHub Actions.

Emotion Detection | Python, BERT, NLP, Hugging Face, Scikit-Learn

- Implemented BERT + Focal Loss multi-label classifier on GoEmotions (58,009 samples, 28 emotions); selected Focal Loss over standard cross-entropy to address severe class imbalance across 28 emotion labels.
- Achieved Micro-F1: 0.52, Macro-F1: 0.35, Hamming Loss: 0.043, outperforming TF-IDF + Logistic Regression baseline via data preprocessing and hyperparameter tuning.

FoodPool | React, Flask, PostgreSQL, SQLAlchemy, JWT, REST API, Pytest, Jest

- Architected a full-stack food ordering platform (React, Flask, PostgreSQL) using a client-server design with a stateless REST API, supporting group polling, real-time collaboration, and recurring orders.
- Documented system architecture and API contracts to onboard 3+ collaborators; automated 200+ tests and established CI/CD workflows, reducing delivery costs by 20–35% for 50+ users.

Mood Muze | DeepFace, OpenCV, NLP, Flask, REST API, JavaScript

- Built a browser-based playlist recommender that detects facial emotion from a live webcam feed and maps 7 emotions to playlists, plus a location mode.
- Ran all inference client-side with locally bundled model weights, so no video leaves the device; increased recommendation relevance by 30% and achieved 90% user satisfaction across 50+ users through iterative requirement-driven UX improvements.